Tensile Workbench — Independent Audit Report

Scientific correctness, specimen attribution, endpoint and yield selection, saved state and exports · audit only, no code changed

Item	Detail
Repository	https://github.com/pKai3/tensile-workbench
Commit audited	`fdcb232c4ff06cd6320db735bb7841b3b50e863c` (2026-10-26, “Standardise saved analysis and plot terminology”)
Audit date	26 October 2026
Environment	Fresh clone in an isolated sandbox; throwaway virtual environment in /tmp (Python 3.12.3, not the required 3.13; numpy 2.5, pandas 3.1, anywidget 0.11; voila/jupyterlab not installed). `TENSILE_WORKBENCH_BACKUP_DIR` pointed to a temporary folder. No research data, personal settings or real outputs were used; git status remained clean. Nothing was committed or pushed.
Priority focus	Validity of curve and averaging workflows; inconsistent behaviour; automatic misattribution or mix-up of specimens and sample sets; logic of automatic YS and failure-endpoint selection.

1. Summary assessment

The code base is carefully structured and unusually well documented. Per-specimen calculations can be traced from original CSV rows to results, the failure-endpoint audit trail is thorough, saved settings are written atomically with conflict detection and verified backups, and landmark averaging behaves exactly as documented. The existing test suite passes on all numerical paths.

However, the green suite does not cover the failure modes that matter most for research results. The audit reproduced four ways the tool can **silently** produce wrong group statistics, plus several cases where a flag is shown but the questionable value is still used:

- **Sample-set misattribution:** one stray CSV or an empty folder named “…exports” in an organisational folder pools all sample groups beneath it into one group (C1).
- **Specimen double counting:** a re-exported duplicate file is loaded as a second specimen and assigned the same Instron row without any check (C2).
- **Endpoint misclassification:** smooth progressive failures with quantised or slightly noisy load are labelled “ISO-style sudden drop (confirmed)”, under-reporting EL by up to ~1 percentage point with no review flag (C3).
- **Unit mis-scaling:** percent strain without a units row is multiplied by 100 for low-ductility specimens; unrecognised stress units and true-stress columns are accepted silently (C5, C6).
- **Flagged but still used:** the fixed 20–50 % UTS elastic window gives large YS errors for work-hardening materials (C4); Instron rows that fail identity verification still feed statistics, reconstruction and Location (C7); excluded values appear as ordinary numbers in the Excel Specimens sheet (C9); unreadable files disappear from analysis while still counted as tested (C10).

Overall: suitable for exploratory analysis with careful review, but group-level results should not be relied on for publication until C1–C3, C5–C7 and C10 are fixed or explicitly checked for each dataset.

2. Findings at a glance

ID	Severity	Evidence	Finding	Primary location
C1	High	Executed	Stray CSV / “exports” folder silently pools sample groups	tensile_selection.py:85–110
C2	High	Executed	Duplicate exports double-counted; same Instron row, no check	tensile_instron.py:178–189; tensile_workbench.py:276

ID	Severity	Evidence	Finding	Primary location
C3	High	Executed	Progressive failure labelled “confirmed sudden drop”; EL under-reported	tensile_fracture.py:275, 351
C4	High (flagged)	Executed	Fixed 20–50 % UTS elastic window biases YS/E for hardening materials	tensile_properties.py:162
C5	Med–High	Executed	Percent strain ×100 when no units row (EL ≤ 4 %); header “(%)” ignored	tensile_core.py:180–209 (204)
C6	Medium	Executed	kPa/GPa/psi and true-stress columns accepted silently	tensile_core.py:100–112, 153–178
C7	Med–High	Executed	Identity-failed Instron rows feed statistics, gauge length, Location	tensile_tables.py:284, 359; tensile_workbench.py:298
C8	Low–Med	Executed	Implausible Location defaults from labels; no count check	tensile_locations.py:15–19
C9	Medium	Executed	Excel Specimens sheet shows excluded values as ordinary numbers	tensile_tables.py:301; tensile_exports.py:18
C10	Medium	Executed	Unreadable CSVs silently dropped but counted as “tested”	tensile_core.py:634; tensile_workbench.py:284

Severity reflects the potential to change published group results without the user noticing. “Flagged” means the tool shows a review message but still uses the value.

3. Confirmed defects

C1	A stray file or folder silently merges sample groups	High
Location	`tensile_selection.discover_sample_groups`, lines 85–110 (rule at 103–104).	
Trigger	A folder becomes a sample group if it directly contains any .csv/.is_tens/.id_tens file, or any subfolder whose name ends in “export(s)”. Its whole subtree is then collected recursively and child folders are never considered as groups.	
Expected	Project A/Demo_Strong and Project A/Demo_Ductile remain two groups; an unrelated file is ignored or flagged.	
Actual	Adding “Project A/specimen list.csv” — or only an empty folder “Project A/Old exports” — produced one group “Project A” containing all six specimens.	
Impact	Two materials are averaged as one sample with no warning. Saved analyses lose their groups; move detection raises nothing because no file moved. The pooling is deliberate for nested runs and is enshrined in `tests/test_group_folders.py::test_instrument_file_or_summary_keeps_nested_runs_in_same_group`, but the rule cannot distinguish runs from nested sample groups.	
Fix	When a candidate group has descendant folders that would themselves qualify as groups, stop and require explicit confirmation (or treat descendants as groups and warn about the parent-level file).	
Regression	Stray CSV in organisational folder; empty “… exports” folder; legitimate nested runs under a summary.	

C2	Duplicate exports are double-counted and share one Instron row	High
Location	`tensile_instron._export_identity` (line 187 accepts both _N and _N_1); `InstronSummaries.match`; no duplicate check in `tensile_workbench._load` (line 276).	
Reproduction	B11.is_tens_Exports containing B11_1.csv, B11_2.csv and B11_2_1.csv (byte-identical to B11_2). Both copies matched row 2 / “Wall_02” with zero check messages.	
Impact	n inflated, mean biased toward the duplicated specimen, SD reduced; tested count and Location plots also affected.	
Fix	Flag identical source SHA-256 within a group and any summary row assigned to more than one curve; exclude extras pending review.	
Regression	Identical _N / _N_1 exports; two different curves claiming one row.	

C3 — Progressive failures labelled “ISO-style sudden drop (confirmed)” — High

Location

`tensile_fracture._detect_uninterrupted_endpoint`, eligibility rule at line 351; noise estimate in `detect_drop_onset`, line 275.

Trigger

Smooth necking to zero load (no sudden drop) with realistic quantisation or noise. The ASTM-style 10 %-of-peak crossing is the expected endpoint.

Actual

Load signal	Selected criterion	Load at endpoint	EL (ref. 29.381 %)	Review flag
Clean	10 % crossing	10 %	29.381 % (correct)	—
Quantised 0.5 MPa	ISO sudden drop, confirmed	26 % of peak	28.316 % (−1.07 pp)	none
Noise 0.2 MPa	ISO sudden drop, confirmed	14 % of peak	29.152 % (−0.23 pp)	none
Noise 0.2 MPa, accelerating tail	ISO sudden drop, confirmed	49 % of peak	29.029 % (−0.81 pp)	none

Cause

A one-quantum decrement after a zero change has an infinite 5× ratio. The selected “drop” equalled the local median decrement and lay 285 readings before the 10 % crossing. The noise floor only guards the numerator, and the MAD-based noise estimate collapses to zero on quantised data.

Impact

Calc EL, toughness, landmark endpoint and reconstructed EL are all affected, and the audit trail records the result as ISO-confirmed.

Fix

Require the candidate decrement to exceed the local decline trend (e.g. median of preceding decrements) by a margin; floor the denominator by noise and an estimated quantisation step; do not label a candidate “confirmed” unless it is significant against the trend.

Regression

The four fixtures above must return the 10 % crossing, alongside the existing sudden-drop tests.

C4 — Fixed 20–50 % UTS elastic window biases YS and E — High (flagged)

Location

`tensile_properties.specimen_calculation`, elastic mask at line 162.

Test

Ramberg–Osgood curves with exact Rp0.2 = 250 MPa and E = 200 GPa, dense sampling.

Result

YS/UTS	Calculated YS	Error	Fitted E	R^2	Flag
0.79	250.0 MPa	0.0 %	200 GPa	1.000	no
0.62	252.8 MPa	+1.1 %	147 GPa	0.974	yes
0.51	273.7 MPa	+9.5 %	49 GPa	0.884	yes
0.38	339.3 MPa	+35.7 %	18 GPa	0.906	yes
≤ 0.24	unresolved	—	11 GPa	—	yes

Impact

Every biased case was caught by the $R^2 < 0.98$ flag, but flagged values still enter tables, plots and landmark means unless the user overrides or excludes them. Relevant to annealed steels, stainless, Al and Cu alloys; acceptable for high YS/UTS alloys such as Ti-6Al-4V.

Fix

Anchor the window to a preliminary yield estimate or use an iterative/maximum-slope window; optionally withhold flagged YS from group means until reviewed.

Regression

Ramberg–Osgood family, YS/UTS 0.3–0.9, known Rp0.2.

C5	Percent strain silently multiplied by 100	Med-High
Location	`tensile_core.compute_strain_percent`, lines 180–209 (heuristic at 204).	
Trigger	No units row and strain median ≤ 2, 90th percentile ≤ 5: the column is assumed to be a fraction. A unit written in the header (e.g. “Strain (%)”) is ignored.	
Reproduction	True EL 3 % → reported 300 % (UE 220 %, E 1.1 GPa); same with header “Strain (%)”. True EL 4 % and 6 % → correct.	
Impact	Result depends on ductility, so specimens within one group can be scaled differently.	
Fix	Parse units from headers; require an explicit unit or show a visible per-file assumption instead of guessing.	

C6	Unrecognised stress units and true-stress columns accepted silently	Medium
Location	`tensile_core.compute_stress_mpa`, lines 153–178 (median rule at 170); column lists at 100–112.	
Reproduction	UTS 475 MPa written in kPa, GPa and psi was reported as 0.4749, 0.4749 and 0.06888 “MPa”. A file with only “True Stress (MPa)” / “True Strain (%)” loaded as engineering data (UTS 575 vs 475 MPa) with no message.	
Fix	Explicit unit conversion table; reject unknown units; exclude or confirm true-stress/true-strain columns.	

C7	Identity-failed Instron rows still feed statistics, reconstruction and Location	Med-High
Location	`tensile_tables.property_tables` (references appended at line 284, summarised at 359); label assignment `tensile_workbench._load` line 298; gauge length in `tensile_gauge.gauge_record`.	
Reproduction	Summary row 2 with UTS 999 MPa against a CSV maximum of 850.45 MPa. The UTS check failed and was shown, yet Instron YS 900 MPa entered the Instron mean (750 MPa, n = 2), the paired Calc–Instron comparison (n = 2, mean difference −100.7 MPa), and label “Wall_09” became the default Location (9). Its gauge length would also be used for reconstruction.	
Fix	Treat failed or ambiguous identity as “Instron values unavailable” for statistics, reconstruction and Location defaults, while keeping them visible for audit.	

C8	Implausible Location defaults from operator labels	Low-Med
Location	`tensile_locations.specimen_number`, line 18.	
Examples	“Ti64 HT 950” → 950; “Build_20261023” → 20261023; “Sample 1 repeat 2” → 2; “Wall-04-2” → 2. No check against the group’s specimen count.	
Fix	Flag defaults above the total count and labels with several numeric tokens.	

C9	Excel Specimens sheet shows excluded values as ordinary numbers	Medium
Location	`tensile_tables.property_tables` line 301 builds rows from raw properties; `tensile_exports.write_workbook` applies no cell marking.	
Reproduction	A specimen set to “AVE unreliable throughout” exports YS 649.29 MPa in a plain cell; only a text column lists excluded properties. Column AVERAGE = 618.4 MPa (n = 5) versus the correct Summary mean 610.6 MPa (n = 4).	
Fix	Export effective values in the main columns and raw values in clearly named raw/excluded columns, or at least apply strikethrough and fill.	
Regression	For every data-use mode, the mean of each exported effective column equals the Summary mean.	

C10	Unreadable CSVs silently dropped but counted as tested	Medium
Location	`tensile_core.read_tensile_csv_table` (line 634) / `load_and_prepare_curve`; `tensile_workbench._load` only prints “[SKIP]” (line 284); counts from `tensile_instron.read_catalogue_csv`.	
Reproduction	Decimal-comma exports (semicolon-separated, or quoted “1,23”) are rejected by the numerical loader, yet both are counted by the tested-specimen catalogue.	
Impact	A group can show “5 tested of [5]” while statistics use n = 4; the dropped file is named only on the console.	
Fix	Report skipped files per group in Specimens/Checks; support or explicitly reject decimal-comma input; derive “tested” from the same loader outcome.	

4. Suspected or latent problems (need further evidence)

Issue	Evidence and risk	Suggested action
Variable “Save all” writes every draft cell	Reproduced in isolation (`tensile_variables.py:300`): after a non-forced refresh, saving reverted two newer saved values the user never edited. Not reachable through the UI paths found, which force-refresh drafts; cross-session edits are caught by file conflict detection.	Save only edited cells; refuse if those cells changed underneath.
Specimen count below tested is silently raised	`tensile_workbench.py:1341` reconciles to the tested count and reports “All variables saved”.	Show an explicit validation message.
Drafts discarded on Reload data	Forced refresh on reload and analysis switch; the latter is documented, the former is not.	Warn before discarding unsaved drafts.
Two curve-preparation paths	YS/UTS/UE use `prepared_indices` (drops decreasing-strain rows, sorts); EL/toughness/curves use `prepared_points` on the endpoint prefix (sorts only). Strain-tracking glitches are sorted into the curve rather than flagged; small numerical effect in the test case.	Single preparation path; flag strain reversals.
Pointwise-average SD uses ddof = 0	`tensile_core.py:753` versus sample SD elsewhere. Not confirmed whether this band is displayed or exported.	Use ddof = 1 or label clearly.
“Time (s)” header without units row	By code reading the time scale becomes NaN, silently disabling time-gap detection.	Parse header units.
Ambiguous representative override	First match used; warning printed to console only.	Surface in UI.
Uniform elongation from a single raw maximum	SD 0.13 pp over ten noise seeds; plateau handling per ISO not verified.	Consider plateau-midpoint or smoothed peak.
Cloud-synced application folder	Lock, .previous.json and temp files sit beside the project file and may sync between machines; backup history is keyed by absolute path. Not tested live.	Keep locks outside synced folders; document.
Backup retention vs autosave rate	20 changed versions may cover only minutes of autosaves; daily checkpoints keep each day’s last version.	Add time-based retention.
Documentation drift	Guide says endpoint method “v9”; code says v10.	Update guide.

5. Checked and found working

- **Landmark averaging with mixed eligibility** (five specimens: use all ×2, exclude shape, after UTS, AVE unreliable). The curve passes exactly through the eligible means (YS 610.641 MPa, UTS 773.140 MPa, failure EL 18.324 %); its maximum lies at the UTS marker; shape uses only the two fully eligible coupons; property counts (YS/UTS/UE/EL = 4/5/4/3, shape n = 2) match the Summary sheet.
- **Population rules** are applied consistently in summary tables, group property charts, strength–EL scatter and specimen plots (code reading; summary verified numerically).

- **Failure elongation is never used for identity.** Identity uses embedded summary, dataset + Instron row number or exact label, verified only by UTS within printed-digit tolerance.
- **Excel precision** is retained in full (e.g. 618.3710246542745); two decimals is a display format only.
- **Excel formula injection** is prevented: headers and specimen names are forced to literal strings (tensile_exports.py lines 72, 107).
- **Local server** binds to 127.0.0.1 with a token.
- **Settings persistence:** atomic temp-file + fsync + replace; exclusive lock; digest and inode/mtime conflict detection; mandatory pre-migration and pre-restore safety copies; checksummed backups with read-back verification; protected snapshots exempt from pruning.
- **Move protection:** duplicate-content and multi-destination moves are blocked as ambiguous; partial-group disappearances are not migrated; nothing transfers without approval.

6. Standards claims

Verified from the published preview and table of contents: ISO 6892-1:2019 is the third edition; Annex A (computer-controlled testing) is *informative*; clauses 17–20 separately define plastic and total extension at maximum force (Ag, Agt), total extension at fracture (At) and elongation after fracture (A).

Implication: the workbench's Calc EL is a total strain at the selected endpoint, i.e. At-like, not A. Plots and exports should label it accordingly.

Not verified (full texts are paywalled): the exact wording of ISO 6892-1:2019 A.3.6 (the >5× / <2 % rule) and ASTM E8/E8M-25 §7.11.3.3 and §7.11.3.4 (10 %-of-peak criterion). The application's citations of these clauses should be treated as unverified claims; the implemented safeguards are application heuristics, as the documentation itself states.

7. Checks performed and limitations

Existing test suite

```
TENSILE_WORKBENCH_BACKUP_DIR=$(mktemp -d) MPLBACKEND=Agg python -m unittest discover -s tests
```

Result	Count	Notes
Run	381	≈120 s
Skipped	3	—
Failures	11	All in test_environment_setup: runtime guard requires Python 3.13 (environmental).
Errors	2	test_browser_launch and test_loading_artwork: voila / nbformat not installed (environmental).
Numerical tests	all pass	None exercise the scenarios in C1–C10.

Adversarial experiments (synthetic, temporary)

Area	Experiment	Outcome
YS selection	Ramberg–Osgood family, exact Rp0.2, dense and coarse sampling	C4
Units	Unitless / "(%)" header strain at EL 3–6 %; mm/mm; με; kPa/GPa/psi/N·mm^{-2}; true-stress-only file	C5, C6
Endpoint	Progressive failure: clean, quantised 0.5/0.05 MPa, noise 0.2/1.0 MPa, accelerating tails; sudden fracture controls	C3
Uniform EL	Ten noise seeds near peak	Suspected list
Curve preparation	Strain-tracking glitch (−2 pp for 60 readings)	Suspected list
Group discovery	Stray CSV and empty "exports" folder in organisational folder	C1

Area	Experiment	Outcome
Attribution	Duplicate _N/_N_1 exports; UTS-mismatched summary row	C2, C7
Location	Label parsing edge cases	C8
Landmark / populations	Full PreviewSession with mixed data-use modes	Working (section 5)
Exports	Table-only Excel export: precision, exclusions, cell formatting	C9; precision OK
Import robustness	Decimal-comma CSVs vs tested-count catalogue	C10
Variable editor	Stale draft vs newer saved values; count below tested	Suspected list

Not verified

- Visual parity between static and Plotly figures, and label rendering.
- Browser-side paste, copy and quoting in the variable table (server-side `paste_cells` validation is rectangular and atomic by code reading).
- Performance and cache behaviour on large datasets and “show all” actions.
- macOS app and Windows launcher behaviour; setup under Python 3.13.
- Live behaviour in cloud-synchronised folders.
- Full standards texts (see section 6).

If a finding needs confirmation on real data, the smallest useful anonymised example is: one group folder with its Instron summary CSV and two or three raw exports, with operator labels replaced and stress/strain columns kept unchanged (to preserve quantisation and noise).

8. Prioritised remediation plan

#	Fix	Findings	Regression checks to add
1	Attribution and population integrity: block silent pooling, flag duplicates and multiply-assigned rows, gate Instron values on verified identity, surface unreadable files	C1, C2, C7, C10	Stray CSV / empty exports folder; identical _N/_N_1 exports; UTS-mismatched row excluded from Instron stats, gauge length and Location; decimal-comma file reported per group
2	Endpoint selection: trend-relative drop test, quantisation-aware noise floor, no “confirmed” label without significance	C3	Progressive-failure fixtures (quantised and noisy) select the 10 % crossing; existing sudden-drop tests still pass
3	Explicit units: parse header units, reject unknown units and true-stress/strain columns	C5, C6	Unitless 3 % EL; “Strain (%)” header; kPa/GPa/psi; true-stress-only file
4	Exports: effective values in main columns, raw values separately marked	C9	Exported effective-column means equal Summary means for every data-use mode
5	Yield window anchored to preliminary yield or iterative; option to withhold flagged YS from means	C4	Ramberg–Osgood family YS/UTS 0.3–0.9 within tolerance of known Rp0.2
6	Hardening: edited-cells-only variable saves, explicit Y < X feedback, Location plausibility, lock files outside synced folders, time-based backup retention, single curve-preparation path	C8 and section 4	Stale-draft save does not revert unedited cells; Location > count flagged; strain-reversal flagged

No application code was changed during this audit. Group 1 is recommended first: those defects change which specimens are counted, and their fixes are small and easy to test.